

ESP32-C3-WROOM02 mini module development board

Main Hardware components

- USB 2.0 type C connector
- LDO (AP211K-3.3) which converts 3.3V required voltage for power on the MCU
- Boot button- to flash the firmware or to go into bootloader mode to flash the firmware
- Reset button- to reset the MCU manually
- External 32.768KHZ crystal oscillator
- Pin header for JTAG/UART
- ESD protection circuit from USB to LDO to protect the USB data line(USBLC6-2SC6)

Working flow of the board

- USB type C connector will be the main power input to the board
- From USB we get the 5 V DC voltage which goes to LDO which convert the 3.3V with protection circuit in between from 5 V to 3.3V
- 3.3V power rail from LDO will direct goes to the power on to ESP32-C3 mini
- Boot button at (GPIO8) is used to put MCU into programming mode
- Reset button (at EN pin) allows to reset the MCU manually
- External crystal 32.768khz provide stable clock cycle
- Pin header for UART/JTAG for debugging mode

USB C connector

- vbus
Power pin got the 5V power from the PC

Cc1 and cc2 pins

Pins used to detect the connection , cable orientation

By keeping pull up or pull down resistors connection at cc1 nad cc2 the cable detected and decided whether i have to act as host (to drive the power to the board) or device (to ask for power from the host)

It is configuration channel pins

This pins tell the host that cable is inserted

So usb plug detection is done by this

By default USB standards the USB type C connector will get the 5V DC from the host (pc or charger)

How can type c verify the orientation of cable detect

In normal insertion cable connects c1

Cc2 not detected

But in flipped insertion

C2- not detected
C1- cable connects

Only 1 cc1 lines is physically connected at a time other one is floating
Host (PC or charger) checks voltage on both pins
When host checks voltage on both pins
If voltage available on c1= normal orientation
If voltage available on c2= flipped orientation

SBU1 and SBU2 pins

Sideband pins
In normal USB2.0 communication this pins are not used and left unconnected
This pin used when USB c used as headphone jack means when this pins carry some audio data

USBLC6-2SC6

This is ESD protection IC used to protect the data lines from USB connector coming
It will protect the data lines from static electricity and voltage spikes

In this 1 and 3 pins are input pins
4 and 5 are output pins
5= vbus
2=gnd

So at 4 and 6 pins getting output USB D+ and USB D- data

LDO (AP2112K-3.3)

This will convert the 5 V which is got from the USB connector and convert it into 3.3v to power on the MCU
As ESP32-C3 module
Ideal current consumption is 20 to 40mA
And active wifi current consumption=300 to 400mA
So decided to have some extra current capacity carrying LDO
AP2112K-3.3 this is 600mA current carrying capacity of LDO
AMS1117-3.3 the dropout voltage =1.3v max
AP2112K-3.3 the dropout voltage is 400mV
So power loss/heat would be less in AP2112K-3.3
Also it has fast response to sudden current peak

MCU

ESP32-C3-WROOM02

VBUS

MCU operates at 3.3V

3.3V rail is the main power supply provider rail

Decoupling capacitor at power line= 10uF and 0.1uF

10uF will filter out low frequency noise

0.1uF will filter out high frequency noise

EPAD pin

This pin should connect with ground pin for good thermal performance

Crystal

MCU has internal crystal oscillator of 40MHz crystal

And internal crystal oscillator (32.768kHz) can be connected with IO0 and IO1

EN pin at pin 2

This pin is used to manually boot the MCU for flashing the firmware or go into bootloader mode

To enter into bootloader mode

Hold boot button (GPIO 9 low)

Press reset (EN pin low)

Release reset

Release boot

So manually push button switch is connected to with this pin

And by default in nominal mode the EN pin is high by pull up resistor

and when switch is pressed it connects with gnd and goes to low

When the switch is pressed the EN pin goes low immediately and when released the switch

EN pin again goes high so this fast switching can create noise in EN pin

So to remove that noise rise delay is added

By $t=RC$

There should be resistor and capacitor in series with the other end of the switch

This RC combination circuit is called debouncing circuit

Also one pull up resistor is connected to the switch to keep the by default switch in high mode

When it is pressed it goes to gnd

USB pins in MCU

IO18 for USB-D-

IO19 for USB-D+

From USB this data lines are connected to MCU io18 and io19 pins

In between from USB to MCU path there is one ESD protection diode which is USBLC6-CS and output of this diode is goes into MCU data lines pins

Strapping pins

This pins are used to decide MCU operating mode (normal flashing mode or bootloader mode)

GPIO2, GPIO8, GPIO9 are for strapping pins

While power on time the MCU checks this strapping pins at that time if pins are high then MCU run into normal mode so it will run the flashed firmware

But if GPIO8, GPIO9 are low= run into bootloader mode

To keep the strapping pins high for normal mode 10kohm pull up resistors are required

TX, RX pins of the MCU

IO20 and IO21 are the TX and RX pins

IO20- RX

IO21-TX

Crystal connection with MCU

There are 2 load capacitors required with external crystal to stabilize the oscillation

And one resistor to help oscillations start easily

Load capacitors are used to stabilize the oscillations.

Load capacitors values are mentioned in the datasheet of crystal oscillator

COMPONENT SELECTION and TECHNICAL SPECIFICATION

USB section

At cc1 and cc2 pins there is 5.1kohm pull down resistors

This pull down resistors used to tell the host that i am the device give me power through USB connector

5.1kohm value this is mentioned in the type c specification

Here why pull down resistors connected what if i connect pull up resistors

If pull up resistors connected at cc1 and cc2 it mean usb will drive the power (it would become host)

So in case when you want to give the power to the board by USB supply then connect pull up resistors at cc1 and cc2 pins

By pull down resistors connection at cc1 and cc2 = i am the device give me the power supply
Asking for power from the host

LDO decoupling capacitors

At vin and vout pin there is decoupling capacitors needed to remove high frequency noise and low frequency noise

Here 4.7uF used

1uf to 4.7uf value range= to remove low frequency noise

0.1uf= to remove high frequency noise

In datasheet 1uF is mentioned for vin and vout

We can use 4.7uf also

BOOT SWITCH AND RESET SWITCH

As the by default the boot and reset switch need to keep at high state so that whenever user want to enter into boot mode by pressing the switch you can do it

So pull up resistors required to keep the switch by default in high state

Note: should keep low value for pull up resistors in the switch so that by weak enough the pin easily goes to low when button is pressed

Here 10kohm is enough to keep the line low

Debounce circuit (R and C) value at switch

R=10kohm

C=0.1uf

$t=RC= 10\text{kohm} \times 0.1\text{uf} = 1\text{ms}$

This will add 1ms of time delay in rise of the EN or reset pin because if it goes immediately high after pressing the switch then noise might be added

This combination of the RC at switch is called debounce circuit

WHY EXTERNAL CRYSTALS NEEDED WHEN MCU HAS INTERNAL CRYSTAL ENOUGH

When internal crystal lacks at stability, precision then external crystal is used

Both load capacitors have same value at crystal connection

And one resistor in parallel